

PDF Tool

Audit #4 — Editor Fidelity Cycle

Live: <https://pdftool4u.duckdns.org> (AWS EC2)
Date: 16 July 2026 · Prepared for: Mukesh Kumar

OVERALL PRODUCTION READINESS

93 / 100

Up from 92. The editor is now the app's strongest feature. The score is capped by operational items untouched since Audit #3 — chiefly the old Render deployment, whose suspension remains unconfirmed.

1. Executive Summary

This cycle was entirely product work, all in the PDF editor. Five releases landed since Audit #3, verified in the commit history and green in CI: text-color matching via pixel sampling, real font-data weight detection, Inter family embedding through fontkit (with granular weights and the Display optical variant for headlines), bold/italic serif and oblique sans support, and corner-handle resizing for text, images and boxes. The owner drove this with adversarial testing on real-world documents (brand-font receipts) — exactly the right way to harden a feature.

The result is a click-to-replace text editor that matches size, weight, color and closest typeface, with optical-size awareness — capability beyond most free web PDF editors and at the practical fidelity ceiling for tools without access to proprietary font files.

The cap on this score is operational, not technical. Every open finding from Audit #3 remains open. In particular, the retired Render deployment was still publicly serving the pre-privacy build at last verification, and its suspension has not been confirmed despite being the standing P0 across two audits. Live-state checks from this audit environment returned cached responses, so current state could not be re-verified here — the owner should confirm directly.

2. Scorecard vs. Audit #3

Area	Score	Δ	Note
Architecture	9.5	=	Unchanged; privacy design remains the differentiator
Security & privacy	9.0	=	No regressions; new font code is read-only and fail-safe
Deployment / DevOps	8.5	=	No movement on Render/Elastic IP/root containers
Code quality	9.0	▲	Font subsystem cleanly layered (schema → matcher → loader → embed) with graceful fallbacks at every level
Features & UX	9.5	▲	Editor fidelity + resize; capped only by proprietary-font ceiling
Testing / CI	7.5	=	CI green throughout; editor/browser code still lacks automated coverage
Performance	8.0	=	Pixel sampling and font embedding add negligible overhead (cached, subset)
Documentation	9.5	=	July 2026 documentation edition produced

3. What Shipped This Cycle

- **Color fidelity:** replacement text samples the dominant glyph color from the rendered canvas (ground truth), with drawing-command recovery as fallback — fixed grey/colored text coming out black.
- **Weight fidelity:** real font objects (PostScript name + bold/black flags) drive matching — fixed bold text coming out regular, since pdf.js internal font ids defeat name regexes.
- **Typeface matching:** geometric/grotesque brand fonts (Uber Move, Circular, Gotham, Roboto, ...) map to embedded Inter — granular weights Medium→Black, InterDisplay at ≥20pt for headline optics; serif bold/italic and sans oblique no longer flatten to regular.

- **Robust degradation:** every layer falls back (unknown weight → nearest standard font; missing font files → Helvetica + log warning) — local dev without fonts-inter keeps working.
- **Resize UX:** bottom-right handle on selected elements; text scales font size, images scale uniformly (signatures keep aspect), boxes resize freely; live preview, undoable.
- **Docs:** July 2026 Technical Documentation superseding the pre-migration edition.

4. Findings

#	Severity	Finding	Status / Recommendation
F1	HIGH	Old Render deployment serving the pre-privacy build. Suspension requested in Audits #2–#3; completion never confirmed. This audit could not re-verify (cached responses).	CARRIED. Suspend both services at dashboard.render.com and confirm. Until verified, treat as live.
F2	MEDIUM	Containers run as root.	CARRIED. Add USER to both Dockerfiles.
F3	LOW	Legacy activity-logging toggle in Settings controls nothing.	CARRIED. Remove or repoint to the browser-local log.
F4	LOW	Lockfile deleted at build time (non-reproducible builds).	CARRIED.
F5	LOW	No E2E coverage; editor fidelity features are manually tested only.	CARRIED; grows in importance as editor complexity grows.
F6	LOW	Elastic IP unconfirmed — stop/start changes the public IP and breaks DNS.	CARRIED.
F7	INFO	No sitemap/robots for the public /help page.	CARRIED.
F8	INFO	Mobile pass not done; new resize handles especially need touch testing.	CARRIED, updated.
F9	INFO	NEW: audit-environment live checks returned cached data; independent verification of live state (Render down, EC2 current build) should be done by the owner in a browser.	Verify manually after each deploy.

5. Verdict & Action Plan

93/100. Product velocity this cycle was excellent and the editor is now genuinely differentiated: privacy-first storage plus fidelity-matched text editing is a combination few free tools offer. But the operational debt column has not moved in two audits, and one item on it is a live contradiction of the product's core promise.

- **P0 (10 minutes, today):** Suspend both Render services and verify in a browser that pdftool-web.onrender.com no longer responds. This is the third audit carrying this item.
- **P1 (this week):** Elastic IP; non-root containers; remove the dead activity toggle.
- **P2:** Mobile/touch pass (resize handles); sitemap + Search Console; Playwright smoke suite for the editor round-trip.
- **P3:** Offline PWA; opt-in AI features (privacy-consistent design); licensed-font drop-in.

